

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	19 February 2026
Team ID	LTVIP2026TMIDS89796
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

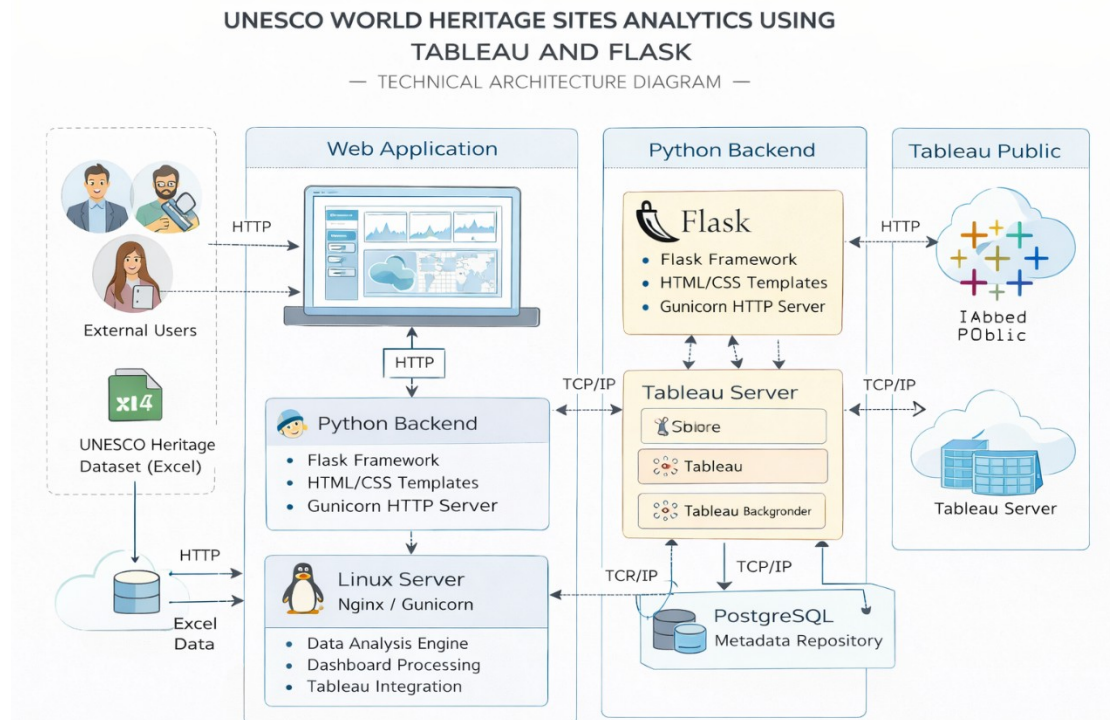


Fig: Technical Architecture

Table 1:

S.No	Component	Description	Technology
1	User Interface	Web interface where users interact with dashboards and project stories	HTML, CSS, JavaScript
2	Application Logic-1	Backend logic to load templates, routing, and server execution	Python (Flask Framework)
3	Application Logic-2	Data visualization and analytics processing	Tableau Desktop
4	Application Logic-3	Dashboard embedding and interactive storytelling	Tableau Public (Embed Integration)

S.No	Component	Description	Technology
5	Database	Heritage dataset used for visualization and analysis	Excel Dataset (UNESCO Data)
6	Cloud Database	Hosting platform for dashboards and analytics	Tableau Public Cloud
7	File Storage	Storage of project files, templates, and static resources	Local File System
8	External API-1	Visualization embedding service used inside web application	Tableau Public Embed API
9	External API-2	Web browser rendering and HTTP communication	HTTP/HTTPS Web Protocol
10	Machine Learning Model	Not implemented in current project (Future Enhancement Scope)	—
11	Infrastructure (Server / Cloud)	Application deployment environment	Local Flask Server (127.0.0.1), Future Cloud Deployment Possible

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The project uses open-source web frameworks to build and run the application interface and backend logic.	Python Flask, HTML, CSS, JavaScript
2.	Security Implementations	Secure embedding of Tableau dashboards, controlled routing in Flask, and safe hosting through Tableau Public.	Flask Routing, HTTPS (Tableau Public), Basic Web Security Practices
3.	Scalable Architecture	The system follows a layered architecture where dashboards, web application, and data source are separated, allowing easy addition of new dashboards.	Flask Framework, Tableau Public Integration
4.	Availability	The application remains available whenever the local Flask server is running and dashboards are accessible through Tableau Public cloud hosting.	Local Flask Server, Tableau Public Cloud
5.	Performance	Lightweight Flask backend and optimized Tableau dashboards ensure smooth loading and responsive interaction for users.	Python Flask, Tableau Visualization Optimization